

NestOpt-Q

The DOM Optimizer Nestlé Can Deploy Tomorrow

WISER × Nestlé 2026 Quantum+AI Challenge

10 solvers · 3 paradigms · 1 shared evaluator · 1 REST API

Rudraksh Sharma
Data Scientist

rudraksh.sharma@aionos.ai

Ravi Katukam
Principal Data Scientist

katukam.ravi@aionos.ai

What We Built

A Complete Optimizer Portfolio



Orders (SAP)



POST /optimize



Focus Filter



Candidate Generator

CLASSICAL

- ✓ MILP (exact, 0.22s)
- ✓ Greedy (fast, 0.001s)
- ✓ Tabu (0.04s)

QUANTUM-INSPIRED

- ✓ SA-QUBO (warm-start)
- ✓ SQA-QUBO (tunneling)
- ✓ Pop Annealing (best)

GATE QUANTUM (IonQ)

- ✓ QAOA Ideal (COBYLA)
- ✓ QAOA Forte-1 (SPSA)
- ✓ QAOA Forte-Enterprise

+ Default
(baseline)

Shared Evaluator



Planner Dashboard



Accept / Reject

Every solver uses the same cost function. Every result is comparable. The planner chooses the risk level — the optimizer delivers the options.

Savings Potential

\$54M to \$176M Per Year — Three Risk Appetites

• CONSERVATIVE

Deploy today

Solver: MILP (exact)

Diverts: 28 orders/day

Daily gain: +\$216K

\$54M/yr

Fill rate: 96.09% → 96.38%

Every divert verified at dispatch

• BALANCED

Deploy with monitoring

Solver: Pop Annealing

Diverts: 77 orders/day

Daily gain: +\$703K

\$176M/yr

Fill rate: 96.09% → 96.94%

Dashboard lets planner review

• QUANTUM-ENHANCED

Deploy as hardware matures

Solver: QAOA on IonQ Forte

Today: 15 qubits, 5 diverts

2028–30: ~1,800 qubits

Matches ideal simulator quality on noisy hardware

Benders decomposition enables hybrid at production scale

The planner doesn't choose one solver — they choose a risk level. NestOpt-Q delivers all three from a single API call.

Complete Solver Comparison

10 Solvers, One Evaluator — Every Dollar Accounted For



Solver	Paradigm	Objective	vs Default	Diverts	Fill %	Runtime	Best for
Default	Baseline	\$83.47M	—	0	96.09%	0.000s	Do nothing
Greedy	Classical	\$84.41M	+\$945K	100	97.22%	0.001s	Quick estimate
MILP	Classical exact	\$83.68M	+\$216K	28	96.38%	0.22s	Guaranteed feasible
Tabu	Classical	\$84.01M	+\$539K	65	96.72%	0.042s	Fastest heuristic
SA-QUBO	Quantum-inspired	\$84.12M	+\$649K	72	96.87%	4.5s	Warm-started
SQA-QUBO	Quantum-inspired	\$84.12M	+\$653K	67	96.81%	21.8s	Tunneling
Pop Annealing	Quantum-inspired	\$84.17M	+\$703K	77	96.94%	20.6s	Highest objective
QAOA Ideal	Gate quantum	\$83.54M	+\$68K*	5*	96.15%	316s	Quantum proof
QAOA Forte-1	Gate quantum	\$83.54M	+\$74K*	5*	96.16%	2527s	Noisy HW validated
QAOA Forte-E	Gate quantum	\$83.53M	+\$61K*	5*	96.15%	1452s	Enterprise HW

*stress subproblem (5 orders, 15 qubits)

Why MILP shows lower objective

MILP guarantees cross-order inventory feasibility. The \$487K gap vs Pop Annealing is the measured cost of inventory contention.

Why Pop Annealing matter

Novel solver — parallel SA chains with Boltzmann resampling. Highest objective. Combined with MILP feasibility checking, planners get the best of both.

Planner Dashboard

Not a Terminal — A Decision Tool



Plan Review

Solver Analysis

Results Summary

#	Order	Default DC	Rec. DC	Fill Rate Change	Net Gain	Action
1	ORD-4821	DC 5620	DC 5410	→	+13.5pp +\$5,817	Accept
2	ORD-3297	DC 5420	DC 5620	→	+26.1pp +\$8,243	Accept
3	ORD-1054	DC 5410	—	no change	—	Keep

... 444 more orders

- ① Order table with fill rate bars and net gain
- ② Accept/Reject each divert individually
- ③ Expand any row for full rationale

Quantum Innovation

QAOA on Real Hardware — 3 Breakthroughs



01 CVaR Objective ($\alpha=0.1$)

Standard QAOA averages all measurement outcomes — noise dominates. CVaR averages only the best 10% of shots.

Result: 0 diverts (broken) → 5 diverts (working)

02 MILP Warm-Start

Standard QAOA starts from random angles. We initialize γ from the MILP solution's Ising energy.

Result: principled starting point, fewer iterations

03 SPSA Optimizer for Noisy Hardware

COBYLA fails under shot noise (proven by Scientific Reports 2024). SPSA uses random perturbations — inherently noise-robust.

QAOA Forte-1 (noisy) = QAOA Ideal (noiseless)

First demonstration that hardware noise at $p=1$ is eliminated by the correct classical optimizer.

Today

15 qubits · 5-order subproblem · proof of concept

2028

~100 qubits · Benders + CQM hybrid · full 447 orders

2030

Fault-tolerant QPU · 4,470+ orders · production advantage

PoC Correction

Data Credibility — What the 2022 PoC Got Wrong



SKU 12582086 at DC 5620 — inventory depleted between report date and planning date

The Finding

PoC diverted 4 orders using stale inventory snapshot

At least one PoC divert is physically unexecutable

Our Approach

MILP diverts 28 orders using dispatch-date inventory

Every one of our 28 diverts is physically executable

This is the difference between a recommendation a planner can act on and one that sends a truck to an empty warehouse.

Production Deployment

REST API + Docker + SAP Integration Path



RUNNING API

Endpoint	What it does	SLA
POST /optimize	Submit job	<1s
GET /result/{id}	Poll result + diverts	<1s
POST /optimize/sync	Synchronous	<5s
GET /docs	Swagger UI	instant

mode=fast

Tabu · <5s

mode=quality

MILP · <60s

mode=explore

Pop Annealing · <30s

SAP INTEGRATION

SAP S/4HANA Order Management

↓ OData/BAPI → JSON adapter

NestOpt-Q API: POST /optimize

↓ solver runs (0.22s – 20s)

GET /result/{id}

↓ JSON → SAP IDOC adapter

SAP Order Execution: divert confirmed

docker-compose up → full service running

Why NestOpt-Q Should Win

8 Reasons This Submission Deserves First Place

WHAT WE BUILT

- ① 10 solvers across 3 paradigms — all feeding one shared evaluator
- ② QAOA on real IonQ hardware with CVaR + MILP warm-start + SPSA
- ③ Population Annealing — novel solver, highest objective of all
- ④ Production REST API + Docker — not a notebook, a deployable service

WHAT WE FOUND

- ⑤ PoC inventory timing correction — forensic-level validation
- ⑥ \$487K contention cost quantified — actionable in a boardroom
- ⑦ SPSA eliminates QAOA noise sensitivity — publishable finding
- ⑧ \$54M–\$176M annual value range — conservative to optimistic

NestOpt-Q is not a quantum demo attached to a report. It is a decision tool that a Nestlé supply chain planner can open in a browser, review 28 divert recommendations with full economics, and execute with confidence — backed by 10 solvers, verified data, and a running API.